Durham University

Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2025	ENGI3371-DE01

Title: (BEng/MEng)

Electrical Engineering 3

Time Allowed:	2 hours
Additional Material provided:	No
Reporting problems during the exam:	Queries about the instructions for this exam (rubric) should be directed to: eng.boe@durham.ac.uk. If you experience any difficulties uploading your exam answers to Gradescope report these immediately to: eng.boe@durham.ac.uk.

Instructions to Candidates:	This is a synchronous, time restricted exam. All submissions must be uploaded to Gradescope as per the published timetable. Answer ALL questions. Please answer each question on a new page. All workings must be shown. This is an online open-book exam. • *The examination attempt must be all your own work. You must not interact with another person (either one-to-one, by telephone, on the internet, in chat rooms, on social media, via smart watches, etc.), or use cheat sites, etc. whilst completing the exam. You must not discuss any examination papers with friends, relatives, current or past students of this or any other University before the end of the exam submission window.* • *Candidates are expected to answer all questions in their own words. Should close paraphrasing, direct quotation be necessary candidates must acknowledge the source of any material used in their answers (this includes text, images, diagrams, charts, tables and/or graphs). Full referencing is not expected, but a brief indication of the source of the material (e.g. name of author) must be given.* • *In line with the Department of Engineering document entitled "Student Guidance on the Use of Generative Artificial Intelligence and Related Technology for the Department of Engineering" policy in assessments, the use of generative artificial intelligence tools and related technologies is strictly prohibited for this exam.*

Revision:	

Section A

Question 1

a) Figure Q1 shows a diode rectifier providing power to a resistive load. Sketch and label the waveforms for v_s (the AC source voltage), v_d (the DC output voltage) and v_1 (the D1 diode bias) for the circuit in two cases: (i) ideal diodes and (ii) non-ideal diodes. Briefly explain the differences in terms of rectifier output voltage.

[30%]

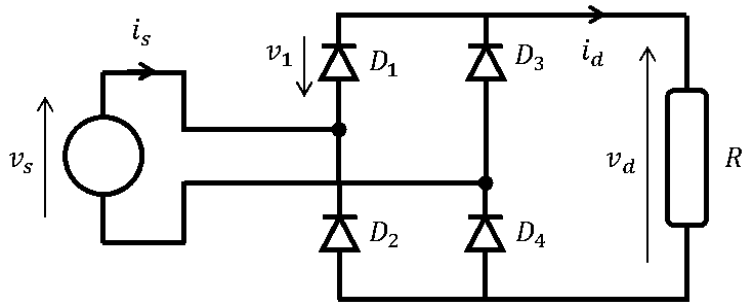


Figure Q1

b) A single-phase full-wave thyristor rectifier can be operated in either discontinuous conduction mode (resistive load) or continuous conduction mode (resistive-inductive load). Using derivations to support your answer, show that the range of possible average output voltages for the two modes of operation for firing angles $0 \leq \alpha \leq \pi$ are as in Table Q1.

[30%]

Table Q1

Conduction Mode	**Average output voltages**	
	$\alpha = 0$	$\alpha = \pi$
Discontinuous	$\frac{2\sqrt{2}}{\pi} V_s$	0
Continuous	$\frac{2\sqrt{2}}{\pi} V_s$	$-\frac{2\sqrt{2}}{\pi} V_s$

c) A bidirectional DC-DC converter is connected to an ideal 48 V DC source (V_{dc}) and driving a permanent magnet DC machine with armature resistance 0.2 Ω. Under rated motoring conditions (V_a = 48 V), the machine runs at 410 rpm, provides 20 Nm torque and draws a current of 25 A. Mechanical losses can be neglected. Two reference equations are below.

$$E_a = k_\omega \omega \qquad\qquad T = k_\omega I_a$$

continued

i) Calculate the machine speed constant (V/rad/s).
ii) Determine the buck switch duty cycle required to provide regenerative braking at 40% of rated torque from rated conditions.
iii) If the machine is operated at rated conditions when the duty cycle is suddenly reduced to 0.2 (20 %), calculate the armature current and braking torque. With reference to electrical and mechanical points, briefly describe why this reduction in duty cycle could be problematic.

[40%]

Question 2

a) Considering the IEEE-recommended circuit model, sketch the power flow diagram of a three-phase induction motor, and:

i) Briefly explain the power loss components of the machine.

ii) What is the difference between the power flow diagram of the IEEE-recommended model and the exact model of the machine?

[30%]

b) A three-phase Y- connected induction motor has the rating values of 400 V, 17.5 HP, six pole, 50 Hz. This motor is used in a factory to drive an industrial fan at rating voltage and frequency, and rating slip of 5.5 %. The torque-speed characteristics of the fan is given as:

$$T_m = 4.87 \times 10^{-3}(\omega_r)^2$$

where ω_r is rating rotor speed in rad/sec.

Determine the rotor speed and torque of the fan when the motor is supplied by the rating voltage and frequency.

[50%]

c) In a grid-connected wind farm:

i) In your own words, briefly explain the differences between a permanent magnet synchronous generator (PMSG) and a doubly fed induction generator (DFIG) configurations.

ii) Sketch a block diagram of both systems and clearly label all components.

iii) Briefly explain the key role of the power electronic converters, and the mechanism of maintaining a constant frequency while wind speed fluctuates.

[20%]

Section B

Question 3

a) Consider an asymmetrical pi-connected circuit, similar to a medium-length transmission line model but with unequal values at the two ends. The sending end has a capacitive admittance Y_1, the series impedance $Z = R + jX$ (resistive and inductive) connects the sending and receiving ends, and the receiving end has a capacitive admittance Y_2. Determine the generalised circuit constants for this circuit.

[25%]

b) For the circuit described above, draw the phasor diagram for a lagging power-factor condition at the load (receiving end). Clearly explain the steps and justify any assumptions you make regarding the magnitudes of the phasors based on realistic system values.

[25%]

c) Now consider a circuit formed by the series connection of two such pi-connected models as described above. Determine the generalised circuit constants for the combined system.

[25%]

d) In the circuit described in part (c), neglect the two outer admittances. For the remaining circuit, draw the corresponding phasor diagram for a lagging power-factor load. Clearly explain the steps and justify any assumptions you make regarding the magnitudes of the phasors based on realistic system values.

[25%]

Question 4

For a three-bus power system:

- Bus 1 is the slack bus with a per-unit voltage of $1.0\angle 0$
- Bus 2 is a PQ bus with a per-unit load of $1.2 + j0.4$
- Bus 3 is a PV bus with a generation power of 1.05 per unit and a voltage setpoint of 1.0 per unit

The per-unit line impedances are: $j0.1$ between buses 1 and 2, $j0.4$ between buses 1 and 3, and $j0.2$ between buses 2 and 3.

a) Derive the bus admittance matrix Y_{bus} for this system.

[25%]

b) Using the Newton-Raphson method, determine the first iteration of the voltages at buses 2 and 3.

[75%]

Formula sheet

Power flow equations:

$$P_i = \sum_{j=1}^{n} |V_i||V_j||Y_{ij}| \cos(\theta_{ij} - \delta_i + \delta_j)$$

$$Q_i = -\sum_{j=1}^{n} |V_i||V_j||Y_{ij}| \sin(\theta_{ij} - \delta_i + \delta_j)$$

Newton-Raphson Power Flow:

$$\begin{bmatrix} \Delta P \\ \Delta Q \end{bmatrix} = \begin{bmatrix} J_1 & J_2 \\ J_3 & J_4 \end{bmatrix} \begin{bmatrix} \Delta \delta \\ \Delta |V| \end{bmatrix}$$

END